

Content

***Lasers:** Characteristics of LASER, Interaction of radiation with matter, requisites of a Laser system. Construction and working of semiconductor lasers. Use of attenuators for single photon sources, Optical modulators and its types, Electro optical modulators (Pockel's effect, Kerr effect), Beam Splitter, Photodetectors – Single Photon Avalanche Diode, Superconducting Nanowire Single Photon Detector, Mach-Zehnder interferometer, numerical problems.*

***Optical Fibers:** Propagation mechanism, Numerical aperture derivation, Modes of propagation. Attenuation in fiber, Discussion of block diagram of Point-to-Point communication, Optical fiber sensor. Numerical problems.*

Introduction:

LASER is an acronym for **L**ight **A**mplification by **S**timulated **E**mission of **R**adiation. It creates a beam of coherent, monochromatic, intense and directional light. Hence laser light is highly organized when compared with ordinary light. This is because the waves of a laser beam move in phase with each other and travel in a narrow path in one direction. Ordinary light spreads out in different directions and is phase incoherent. Lasers are widely used in Engineering, Medicine, Defence, Entertainment, Communication, reading the bar code, cutting and welding metals, displays in light shows, printing documents, guiding missile and so on.

Characteristics of Laser beam

- **Directionality:** The orientation of the mirrors to the cavity axis and the fact that the laser emits photons in a particular direction, ensures that laser output is highly directional and non divergent.
- **Monochromaticity:** The laser beam is characterized by a high degree of mono-chromaticity (single wavelength or frequency) than any other conventional monochromatic sources of light.
- **Coherence:** The degree of coherence of a laser beam is very high than the other sources. The light from the laser source consists of wave trains that are phase locked (identical in phase).

- **High Intensity:** The laser beam is highly intense. Since wave trains are added in phase and hence amplitudes are added. Laser light emits as a narrow beam and its energy is concentrated in a small region and is highly intense and bright.
- **Focus ability:** Since the laser is highly monochromatic, it can be focused so sharply by a lens that the diameter of the spot will be close to the wavelength of the focused light. In actual cases it is much larger than the wavelength (due to non monochromaticity)

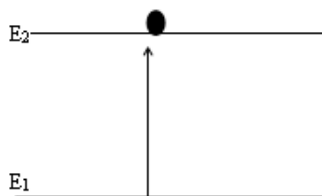
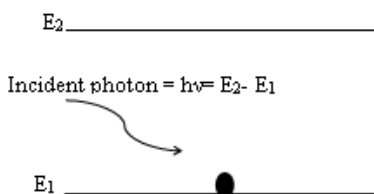
Basic principles: Interaction of Radiation with Matter

Production of laser light is a consequence of interaction of radiation with a quantum system characterized by a set of discrete energy levels. It leads to transition of the quantum system from one state to another state, when it receives or releases an amount of energy equal to the energy difference between those two states.

A two energy level system with lower energy level E_1 and higher energy level E_2 is used to represent an atom. It interacts with photons of energy $E_2 - E_1 = h\nu$ by either absorbing or emitting them. These three types of interactions possible are : Induced Absorption, Spontaneous emission, Stimulated emission.

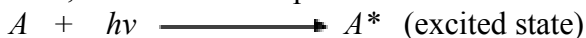
Induced Absorption:

Excitation of atoms by the absorption of photons is called induced absorption.



An atom in the lower energy state E_1 absorbs the incident photon of energy $E_2 - E_1$ and goes to the excited state E_2 .

For an atom A, stimulated absorption transition can be expressed as



The number of absorption transitions per second per unit volume occurring in the material at any instant of time will be proportional to

- (i) The number of atoms in the ground state N_1
- (ii) Energy density of the incident radiation ($U\nu$)

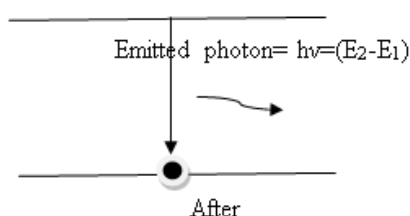
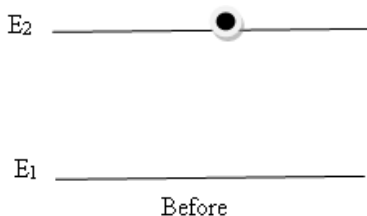
Thus the rate of induced absorption = $B_{12}U\nu N_1$

Here B_{12} is a proportionality constant called Einstein co-efficient of absorption.

Spontaneous Emission:

Here an atom which is at higher energy state E_2 may spontaneously return to the lower energy state E_1 by emitting a single photon of energy $(E_2 - E_1) = h\nu$.

Spontaneous transition can be expressed as



The number of spontaneous transitions per second, per unit volume depends on the number of atoms N_2 in the excited state.

Therefore, the rate of spontaneous emission = $A_{21}N_2$

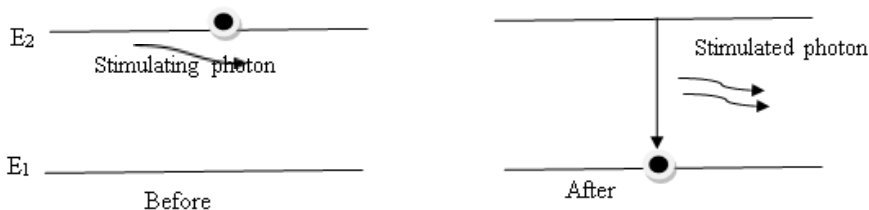
Where A_{21} is Einstein co-efficient of spontaneous emission.

The process has no control from outside. The instant of transition, directions of emission of photons, phases of the photons and

their polarization states are random quantities. There will not be any correlation among the parameters of the innumerable photons emitted spontaneously by the assembly of atoms in the medium. Therefore the light generated by the source will be *incoherent*.

Stimulated Emission:

A new mechanism of photon emission was first predicted by Einstein in 1916 in which an atom in the excited state is induced by a photon of energy $h\nu = E_2 - E_1$ to transit to the ground state by emitting a second photon which is in phase and in the same direction as the original photon. The process is known as induced or stimulated emission of radiation. A single incident photon becomes 2 identical photons by this process. This process can be expressed as :



The number of stimulated transitions per sec per unit volume in the material is proportional to

- (i) The number of atoms in the excited state N_2
- (ii) Energy density of the incident radiation ($U\nu$)

$$\text{Rate of stimulated emission} = B_{21} U\nu N_2$$

B_{21} is the Einstein co-efficient of stimulated emission.

The process of stimulated emission has the following properties.

- I. The emitted photon is identical to the incident photon in all respects. It has the same frequency; phase, direction and same state of polarization.
- II. The process can be controlled externally.
- III. Stimulated emission is responsible for the lasing action.

Some basic definitions

1. Atomic system:

It is a system of atoms or molecules having discrete energy levels.

2. Energy density

The energy density U_v refers to the total energy in the radiation field per unit volume per unit frequency due to photons. It is given by the Plank's distribution law

$$U_v = \frac{8\pi h\nu^3}{c^3} \left[\frac{1}{e^{\frac{h\nu}{kT}} - 1} \right]$$

3. Population

It is the number of atoms per unit volume in a given energy state.

4. Boltzmann factor

The Boltzmann factor $e^{-\frac{E}{kT}}$ is a statistical weight that determines the probability of a system in equilibrium being in a state with energy E at temperature T . It shows that higher energy states are less likely to be occupied. It is the ratio between the population of atoms in the higher energy state N_2 to the lower energy state N_1 at temperature T is given by $\frac{N_2}{N_1} = e^{-\frac{h\nu}{kT}}$, where $h\nu = E_2 - E_1$

Population inversion

Instead of the usual condition of $N_2 < N_1$, if we have the reverse condition, namely $N_2 > N_1$, it is called **population inversion**.

Expression for energy density U_v in terms of Einstein coefficients:

Let us consider the two level atomic system to be in thermal equilibrium with the radiation field, which means that the energy

density is constant with time. This is possible only if the rate of photons absorbed by the system equals the rate of photons emitted by it. Equating the rate of induced absorption to the sum of rate of spontaneous and stimulated emission, we get :

$$B_{12}N_1U_v = A_{21}N_2 + B_{21}N_2U_v \quad \text{or} \quad U_v (B_{12}N_1 - B_{21}N_2) = A_{21}N_2$$

$$U_v = \frac{A_{21}N_2}{B_{12}N_1 - B_{21}N_2}$$

By rearranging the above equation, we get

$$U_v = \frac{A_{21}}{B_{21}} \left[\frac{1}{\frac{B_{12}N_1}{B_{21}N_2} - 1} \right] \quad \text{----- (1)}$$

In a state of thermal equilibrium, $\frac{N_1}{N_2} = e^{\frac{h\nu}{kT}}$

∴ Equation (1) becomes,

$$U_v = \frac{A_{21}}{B_{21}} \left[\frac{1}{\frac{B_{12}}{B_{21}} e^{\frac{h\nu}{kT}} - 1} \right] \quad \text{----- (2)}$$

According to Planck's law of black body radiation, the U_v is given by

$$U_v = \frac{8\pi h\nu^3}{c^3} \left[\frac{1}{e^{\frac{h\nu}{kT}} - 1} \right] \quad \text{----- (3)}$$

Now comparing the equation (2) and (3) term by term on the basis of positional identity we have

$$\frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3} \quad \text{and} \quad \frac{B_{12}}{B_{21}} = 1$$

The fact that $B_{12} = B_{21}$ implies that the probability of induced absorption equals the probability of stimulated emission, though the rates are not. Due to this identity the subscripts could be dropped, and A_{21} and B_{21} can be simply represented as A and B and equation (3) can be rewritten as:

$$U\gamma = \frac{A}{B} \frac{1}{\frac{h\nu}{e^{kT}} - 1}$$

Conditions for light amplification:-

Conditions for laser emission can be studied by taking the ratios of rate of stimulated emission to spontaneous emission and rate of stimulated emission to absorption.

At thermal equilibrium,

$$\frac{\text{Rate of Stimulated Emission}}{\text{Rate of Spontaneous Emission}} = \frac{B_{21}N_2U_\nu}{A_{21}N_2} = \frac{B_{21}U_\nu}{A_{21}}$$

$B_{21}/A_{21} = C^3 / 8\pi h\nu^3 = \text{constant}$. Thus, to enhance the rate of stimulated emission over spontaneous emission, U_ν should be made high.

$$\frac{\text{Rate of Stimulated Emission}}{\text{Rate of Induced absorption}} = \frac{B_{21}N_2U_\nu}{B_{12}N_1U_\nu} = \frac{B_{21}N_2}{B_{12}N_1}$$

Since $B_{12}/B_{21} = 1$, the rate of stimulated emission dominates over the rate of absorption only if we have population inversion. The medium acts as a Gain medium (amplifies light), and otherwise attenuates light.

Requisites of a laser system:

The essential components of a laser are

Active medium : to support population inversion.

Pumping mechanism to excite the atoms to higher energy levels.

Population inversion : to ensure that the medium amplifies light.

Metastable state: to ensure that population inversion is achieved.

Optical cavity/resonator: To ensure directionality, monochromaticity.

Active medium: It is the medium in which light amplification takes place. The medium may be solid, liquid or a gas. Only a small fraction of atoms act as active centres by having population inversion.

Pumping Mechanism: To achieve the population inversion in the active medium, the atoms are to be raised to the excited state. The process of supplying energy to the medium to transfer the atoms to higher energy is called **pumping**. Some pumping mechanisms are:

Optical pumping: It employs a suitable light source for excitation of desired atoms. It is used in solid state lasers like Ruby laser and Nd:YAG laser.

Electric discharge: An electric field can ionize in the medium and raise it to the excited state. It is used in gas lasers (ex: Ar^+ laser).

Inelastic atom-atom collision : In this method a combination of two types of gases are used, say A and B. During electric discharge A atoms get excited and they now collide with B atoms so that B goes to excited state. This technique is used in gas lasers (ex: He-Ne laser).

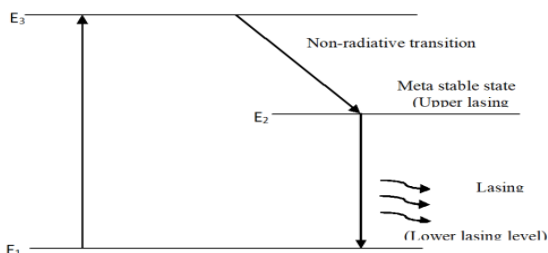
Direct conversion : Electrical bias can be directly converted into light energy. It is used in semiconductor lasers (ex: GaAs laser).

Population Inversion and Meta stable state:

To amplify stimulated emission, we need population inversion. A metastable state is essential for population inversion because it provides a "waiting room" for electrons, allowing them to accumulate in an excited energy level long enough (**10^{-6}s to 10^{-3}s**), compared to 10^{-9}s for normal excited states, for stimulated emission to dominate over spontaneous. This intermediate state is called a **metastable state**.

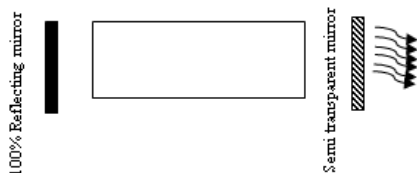
Principle of pumping scheme:

Consider three energy levels E_1 , E_2 and E_3 of a quantum system of which the **level E_2 is a metastable state**. The atoms are excited from E_1 to E_3 state by pumping. The atoms in state E_3 undergo downward transition to either E_1 or E_2 states rapidly. The atoms that transition to level E_2 stay there for a long interval of time, increasing the population of E_2 rapidly and making it higher than even the ground state population in a short period of time. Thus Population inversion is achieved between E_1 and E_2 . The stimulated emission from E_2 to E_1 is the lasing transition.



Optical resonator:

- An optical resonator generally consists of two plane mirrors, with the active material placed in between them. One of the mirrors is semi-transparent while the other one is 100% reflecting. The mirrors are set normal to the axis of the active medium and parallel to each other.



- The optical resonant cavity provides the selectivity of photon states by confining the possible direction of photon propagation, as a result lasing action occurs in this direction.

- The distance between the mirrors is an important parameter as it chooses the wavelength of the photons. Suppose a photon is travelling between two reflectors, it undergoes reflection at the mirror kept at the other end .the reflected wave superposes on the incident wave and forms stationary wave such that the length L of the cavity is given by $L = n \frac{\lambda}{2}$. Hence $\lambda = \frac{2L}{n}$

Where, L is the distance between the mirrors, λ is the wavelength of the photon, n is the integral multiple of half wavelength. The wavelengths satisfying the above condition are only amplified. Hence the cavity is also called resonant cavity.

Role of the optical resonator:

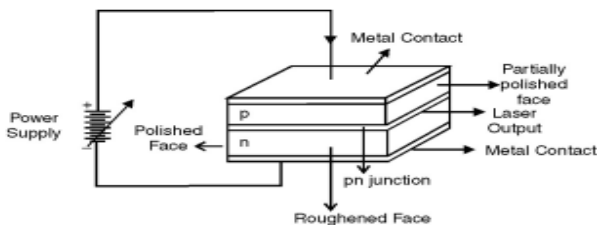
- Provides positive feedback of photons into the active medium to sustain stimulated emission and hence acts as a **source and amplifier of light**. The photon density (U_v) builds up to a very high value through repeated reflections of photons by mirrors.
- Amplifies only the direction of stimulated photons which are travelling parallel to the axis of the optical resonator and normal to the plane of mirrors. Hence output laser light is **highly directional**.
- Selects and amplifies only certain frequencies of stimulated photons which obeys condition $\lambda = \frac{2L}{n}$ and emits **monochromatic light**.

Semiconductor diode Laser

A semiconductor diode laser is a forward biased p-n junction. The first semiconductor lasers were made by R.N. Hall and his co-workers in 1962 using a direct band gap semiconductor material Gallium Arsenide (GaAs). It is portable and used in Photonics, Quantum communications, fibre optics, CD players, CD-ROM drives, optical reading, high speed laser printing etc.

Principle: The semiconductor laser is based on the principle of electron-hole recombination in heavily doped forward biased p-n junctions with a high applied current density. They are usually made

of direct band gap semiconductors such as GaAs, CdSe, GaAlAs etc. Population inversion condition is created in the active region and stimulated emission of light occurs.



Construction: The semiconductor laser consists of heavily doped n - GaAs and p - GaAs. The diode chip is $500\mu m$ long and about $100\mu m$ wide and thick. The junction (active region) lies in a horizontal plane through the centre. The top and bottom faces are metallized and ohmic contacts are provided to pass the current through the diode. The front and rear faces are polished, parallel to each other and perpendicular to the plane of the junction. The polished faces constitute the Fabry- perot resonator. The other two sides of the diode are roughened to prevent lasing action in that direction. The diode is forward biased, with a voltage nearly equal to the band gap voltage of the semiconductor used.

Working:

- The energy band diagram of a **heavily doped p-n junction** is shown in figure (a). Due to heavy doping on both the n-side and p-side, the donor/acceptor bands broaden and the corresponding Fermi levels extend into conduction/valence bands respectively. At thermal equilibrium and zero biasing, the Fermi levels are aligned across the junction as shown in the figure (a) below.
- Electrons occupy the portion of the conduction band below the Fermi level between E_F and E_C while holes occupy the portion of the valence band above E_F and E_V .

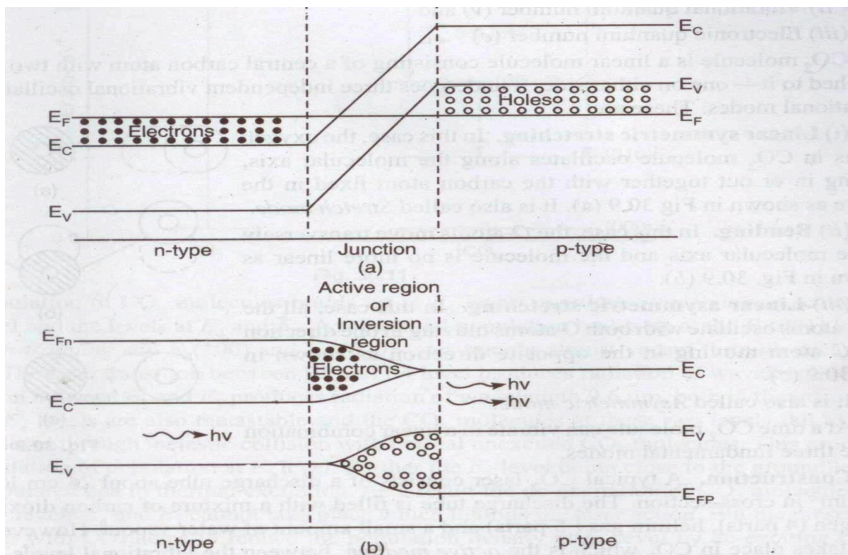


Fig (a) and (b): E-k diagram of a heavily doped p-n junction

- On applying forward bias, the barrier potential gets lowered and electrons from the n region and holes from the p region are injected into the junction. This is the equivalent of pumping in a p-n junction. The uniform Fermi energy splits into two Quasi Fermi energies, E_{Fn} and E_{Fp} , $eV = \Delta E_F = E_{Fn} - E_{Fp}$. Biasing breaks thermal equilibrium, since the populations of electrons and holes in the conduction and valence band respectively change independently.
- For small forward bias, even lower than the band gap, the electron holes recombine and emit spontaneously, non-coherently and in all directions, thus acting like a light emitting diode (LED). At finite temperature, carrier energies are distributed statistically obeying Fermi Dirac distribution. So even before the bias reaches the nominal band-gap voltage, some electrons occupy higher-energy states, some holes occupy lower-energy states and overlap exists between electron and hole populations, allowing for radiative recombinations.

- Once forward bias voltage V increases beyond $eV = E_g$, the system acts as a “Gain Medium” (can amplify light). The concentration of electrons and holes in the conduction and valence band respectively, reaches very high values in the active region, giving rise to the condition of population inversion and triggering stimulated emission. This narrow region in which population inversion takes place is known as ‘**Inversion region**’ or ‘**Active region**’.
- The initial bunch of spontaneously emitted photons stimulate the production of coherent radiation giving rise to laser action. GaAs laser emits at a wavelength of $9000\text{\AA}$ in the infrared region.

Characteristics of GaAs Laser

TITLE	Description
Type	Solid State Semiconductor laser.
Active Medium	The active medium of a semiconductor laser is a p-n junction.
Pumping Method	Direct conversion method.
Power output	The power output from this laser is 1mW.
Nature of output	Continuous-wave or pulsed output.
Wavelength of output	8300 to $9000\text{\AA}$.

Advantages:

1. It is simple, compact and easy to fabricate with high efficiency.
2. The laser output can be controlled by the junction current.
3. It needs lesser power than ruby and CO_2 laser.
4. It can have both continuous or pulsed output.

Disadvantages:

1. The output beam angle has a large divergence from 5° to 15° .
2. The mono-chromaticity is poorer than other types of laser
3. Threshold current density is very large (400A/mm^2).
4. It has poor coherence and stability with low spectral purity.

Optical Attenuators as single photon sources

Optical attenuators are used to attenuate a light beam (reduce its optical power). The amount of attenuation is specified in terms of decibels. Attenuators can have fixed value, wavelength dependence or can be tunable. If the photon flux is reduced dramatically to only 1 photon per time interval, it behaves as a **Single Photon Source**. In general attenuators should not have any effect on Transverse Beam Profile. They can have strong or weak dependency on the Wavelength and Polarization state of light. Neutral density filters are made for weak dependence.

Types of Optical Attenuators based on attenuation mechanism

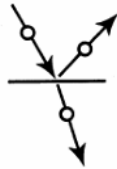
1. Attenuators based on **Optical absorption**

Optical absorption is seen in doped glasses (absorbing filters/plates), where the type and concentration of the dopant can be used to control the amount of absorption in a certain wavelength range. They are low-cost, occupy less volume and have low polarization dependence. The removed optical power is converted into heat, hence thermal effects cause distortions of the beam profile or damage of the attenuator at high power. For intense laser pulses, for some time during and after the pulse, it becomes useless.

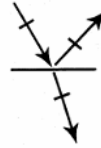
2. Attenuators based on **Fresnel Reflection**

Fresnel reflection describes the reflection of light at the interface of two materials. The amount of light reflected depends on the angle of incidence and the polarization of the light, which defines the orientation of the electric field vector in relation to the plane of incidence. The plane of incidence is a theoretical, 2D flat surface that

contains the incident ray, the normal and the reflected ray. **S-polarized light** has its electric field perpendicular to the plane of incidence, while p-polarized light has its electric field parallel to the plane of incidence.



Perpendicular (S) polarization sticks up out of the plane of incidence.



Parallel (P) polarization is parallel to the plane of incidence.

The reflection and transmission coefficients are given by Fresnel's equations, based on the conservation of light energy. For S-polarized light, the reflection and transmission coefficient (r, t) are given by:

$$r = \frac{n_1 \cos \theta_i - n_2 \cos \theta_t}{n_1 \cos \theta_i + n_2 \cos \theta_t} \quad t = \frac{2n_1 \cos \theta_i}{n_1 \cos \theta_i + n_2 \cos \theta_t} \quad \text{---(1)}$$

Where n_1 and n_2 are the refractive indices of the first and second media, respectively, θ_i/θ_t is the angle of incidence/transmission ($n_1 \sin \theta_i = n_2 \sin \theta_t$). For P-polarized light, the equations are given below:

$$r = \frac{n_2 \cos \theta_i - n_1 \cos \theta_t}{n_2 \cos \theta_i + n_1 \cos \theta_t} \quad t = \frac{2n_1 \cos \theta_i}{n_2 \cos \theta_i + n_1 \cos \theta_t} \quad \text{--- (2)}$$

As the incident angle reaches the **Brewster angle** given by the condition $\tan \theta_B = n_2/n_1$, the reflection co-efficient of the P-polarized light is zero, highlighting its attenuation of P-polarized light.

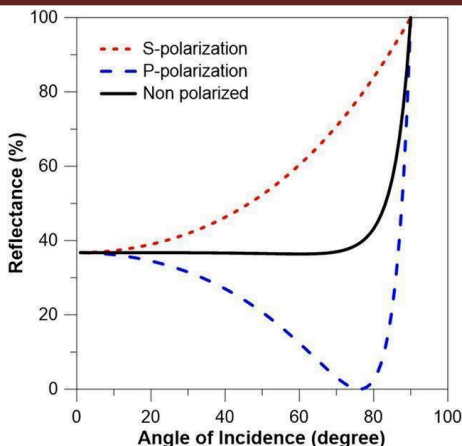


Fig: S and P polarization reflections in Air-Silicon interface

As can be seen in the figure above, the larger the angle of incidence, the higher is the difference in reflectance of s and p type polarization. **A strongly attenuated reflected P-polarized beam can be obtained with an angle of incidence near Brewster's angle, but is unsuitable,** as the presence of even a small amount of S-polarized light in the input beam will contaminate it. To remove polarization dependency of attenuation, one uses multiple reflections at small angles in sequence. Alternatively, a highly reflecting coating is used to obtain a high degree of attenuation for the **transmitted light**.

Materials used for Fiber-optic photon attenuators: Optical fibers doped with rare earth elements like erbium or ytterbium which absorb light at specific wavelengths, thin-film coatings (silicon or gold) deposited on fiber ends, absorb and reflect light, reducing the amount that passes through. Fused fiber couplers and metal-ion doped optical terminators are also used.

Optical Modulators and Their Types

An **optical modulator** is a device that modulates one or more properties of a light wave, like **intensity, phase, frequency,**

polarization, by changing the refractive index, polarization state, absorption coefficient, or optical path length of the light propagating medium, using an external control signal. They are essential components in optical communication, signal processing, laser technology, integrated photonics, quantum optics and Quantum state manipulation. They are classified based on: 1. Modulated quantity: Amplitude/Intensity, Phase, Frequency, Polarization, 2. Physical mechanism used for modulation (Electro-optical effect, Magneto optical effect, Thermo-optical), 3. Configuration (Bulk modulators, Integrated waveguide modulators, Interferometric modulators like Mach–Zehnder)

We will focus on two important types of electro optical modulation called the Pockels effect and Kerr effect.

Electro-Optical Modulation: Pockels and Kerr Effect

Electro-optical modulators exploit the **change in refractive index of a material under an applied electric field**. If the leading order correction in refractive index (dominant term) is proportional to E it is called Pockel's effect, while if the term is proportional to E^2 , it is called Kerr effect.

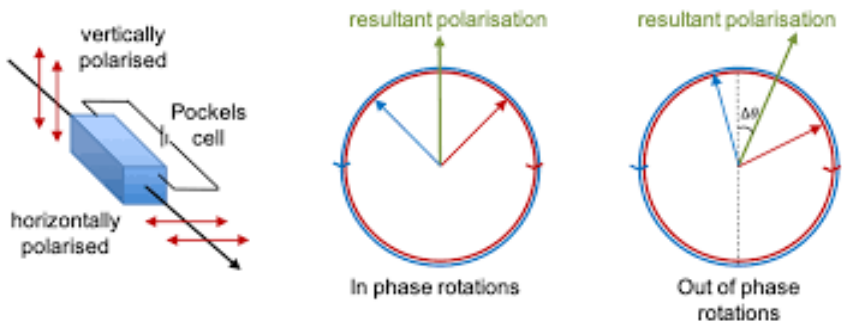
Pockels Effect is a linear Electro-Optic Effect, where the change in refractive index of a material is proportional to applied electric field. It only occurs in **non-centrosymmetric crystals** (crystals with no inversion symmetry), like **LiNbO₃, KDP, BBO**. It is used for enabling ultrafast control of light phase, polarization, and intensity. The Refractive index change is expressed as $\Delta n \propto E$.

The response time is very fast in the GHz range. Switching times can be **nanoseconds or faster** since the mechanism is purely electronic (no moving parts). It requires relatively low operating voltage and finds applications in High-speed optical modulators, Quantum switching in lasers, Optical switches.

Physical Origin of Pockels Effect :

In non centro-symmetric crystals, the internal charge distribution is already asymmetric. Applying an electric field **distorts the electron cloud further**, thus modifying the **dielectric constant** and in turn the **refractive index**, in a **linear** fashion.

Applying an electric field on the Pockels crystal **rotates optical axes**, changes **principal refractive indices** (the three distinct refractive indices (n_x, n_y, n_z) **measured along the three mutually perpendicular principal optical axes** and induces or modifies **birefringence** (refractive index depends on the polarization and propagation direction of light).



Phase Shift Mechanism of Pockels Crystal :

If light travels through length L , $\Delta\phi = (2\pi/\lambda) \Delta n L$

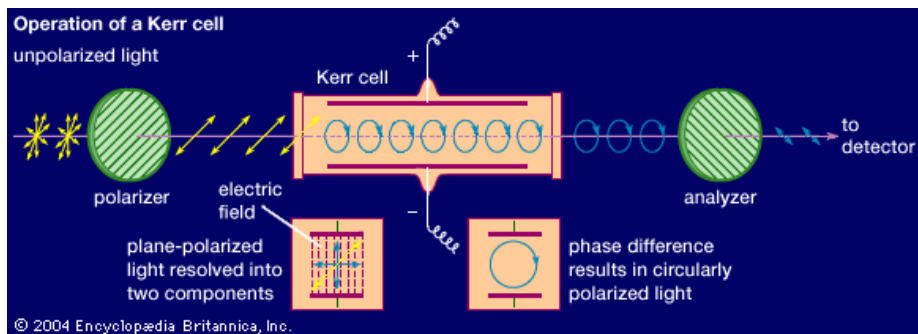
Since $\Delta n \propto E \Rightarrow \Delta\phi \propto V$ (where V is the applied voltage), we see that Pockels cells are perfect **voltage-controlled phase retarders**.

The key applications of Pockels effect is given in the table below :

Device	Role of Pockels effect
Electro-optic modulators	Modulate light phase/intensity
Pockels cells	Fast optical shutters
Q-switching in lasers	Generate giant pulses
Beam deflectors	Voltage-controlled steering
Polarization control	Fast polarization rotators

Common crystals that exhibit strong Pockels effect are : Lithium Niobate which has the strongest Pockels effect and is therefore most widely used, Potassium Di hydrogen Phosphate, which is used in high power Lasers, Beta Barium Borate which is used in Non linear optics, electro-optical switching and Gain amplifiers which is used in integrated and quantum photonics.

Kerr Effect (Quadratic Electro-Optic Effect)



If a crystal has **inversion symmetry** (centro-symmetric) , then since there should be no change in the physical properties as $E \rightarrow -E$, the physical property becomes an even function of E . Hence, there should

not be any linear power of E term. The lowest E dependent term that can survive goes as E^2 . This quadratic electro-optic effect is called the Kerr effect, where $\Delta n \propto E^2$. It is observed in all materials including liquids (e.g., nitrobenzene). It requires **very high electric fields** and is slower and less efficient than Pockels effect. Unlike Pockels effect which comes from structural asymmetry, Kerr effect comes from **intrinsic nonlinear polarization** (P depends on higher powers of E).

Electro-Optic Kerr Effect in DC field: For a static electric field E, the change in refractive index is

$$\Delta n = \frac{1}{2} n^3 K E^2$$

n = refractive index, K = Kerr constant (material dependent, has units of m/V^2). This change in the refractive index causes **Kerr birefringence**. Comparison between Pockels effect and Kerr effect.

Feature	Pockels	Kerr
Relation	Linear	Quadratic
Material	Non Centrosymmetric	All materials
Voltage	Low	High
Speed	Very high	Moderate

Beam Splitter

A **beam splitter** is an optical device that divides an incident light beam into **two or more beams** with a) defined intensity ratios like in an **Amplitude Beam Splitter** which splits light based on intensity (e.g., 50:50) or b) into orthogonal polarization states (**Polarizing Beam Splitter**).

Working Principle : The intensity ratio is precisely controlled by applying specialized thin-film dielectric coatings to the optical surface. Dielectric coatings consist of multiple alternating layers of materials

with high and low refractive indices. The thickness of these layers is optimized to maximize or minimize reflection at specific wavelengths. Metallic coatings (made of aluminum or silver) are avoided since they have high energy loss and introduce polarization dependence. High precision applications require coatings designed to minimize the polarization dependent features, ensuring the two beams maintain identical spectral and polarization profiles. The transmission and reflection ratios are governed by **Fresnel equations**, discussed before.

Applications : It finds applications in Interferometers (Mach–Zehnder, Michelson), Quantum optics (single-photon experiments), Optical instrumentation, Laser guidance systems.

Photodetectors

Photo detectors are used to detect the optical power of incident light (photons). It finds wide application in areas like scientific research, telecommunication, quantum computing, medical diagnostics, photonic sensors and myriads of other consumer based applications. There are different types of photo detectors like Photo multiplier tubes (PMT), charged coupled device (CCD), **Photo diodes**, Photo conductors (LDR) and a few more. Out of these various types, we will focus only on the **Photodiode**. The photo diodes are used to build Avalanche Photo detectors (APD) and SPAD (single photon avalanche diode).

Photodiode: It is essentially a reverse biased p-n junction that converts light energy into an electric current. When light is incident on the active region of a reverse biased p-n junction, electron-hole pairs are created, which are then rapidly swept away in opposite directions due to the internal field generated due to biasing. The electrons/holes move towards the n-type/p-type respectively, forming electron and hole current which constitutes the photocurrent. By measuring the photocurrent, the power of the incident light signal is measured. A normal reverse biased p-n junction generates very low photocurrent

and is hence incapable of detecting very low optical signals. But an Avalanche Photo Diode (APD) can measure very weak signals also.

Avalanche Photodiode (APD)

When the p-n junction is operated at higher reverse bias voltage (typically greater than 6V), an avalanche breakdown takes place within the junction. For diodes with lower breakdown thresholds, usually Zener breakdown occurs. In avalanche breakdown, the current increases dramatically due to carrier multiplication.

Construction and working of APD:

APD is made up of 4 layers p^+ , i, p, n^+ which refer to highly doped p type, intrinsic, nominally doped p type and heavily doped n type semiconductor respectively. The 4 layered structure separates the function of light absorption from the function of avalanche gain (multiplication). This separation is crucial for achieving high internal gain along with efficient collection of photo-generated carriers.

1. The p^+ and n^+ Layers (Heavily Doped Contacts) act as the Anode and Cathode respectively. They provide efficient charge collection at the electrodes and define the outer boundaries of the device.
2. Ultrawide Intrinsic or undoped Layer : It acts as the photon absorption region. Being undoped, its intrinsic region is very wide which maximizes the absorption of incident photons, particularly for long wavelength, near infrared light, generating electron-hole pairs. The large intrinsic layer increases the separation between the p^+ and n^+ layers thereby decreasing the p-n junction Capacitance ($C = \epsilon A/d$), thus allowing it to work more efficiently at higher frequency. The frequency or speed of response of the APD is inversely proportional to

the product of the Load resistance (R) and the junction capacitance C and is given by $f = \frac{1}{2\pi RC}$

3. p Layer (Lightly Doped) : This region is lightly doped to allow a high, uniform electric field to exist across it without inducing premature breakdown. As photogenerated carriers (usually electrons) from the i region drift into this high-field p-region, they gain enough energy to trigger the avalanche multiplication effect via impact ionization. It thus acts as the multiplication region.

Advantages of APD: They have high Sensitivity, high Gain > 100 , and low noise by isolating the multiplication zone. Only the carrier with the higher ionization capability (typically electrons) enters the multiplication zone, which suppresses noise. They have Low Operating Voltages and high frequency of operation.

Single-Photon Detectors

Single-photon detectors are devices capable of detecting **individual photons**, essential in **quantum communication, quantum computing, and low-light imaging**. Key performance metrics are :

a) **Photon Detection efficiency (PDE):** It is the probability that an incident photon on a single-photon detector is successfully registered as a count. It ranges from 10% to over 90% depending on technology (e.g., SNSPDs $>90\%$, Silicon SPADs $\sim 60\%$). It is key for quantifying sensitivity, called Quantum Efficiency (QE).

b) **Dark count rate (DCR) :** The dark count rate (DCR) of an Avalanche Photodiode (APD) is the frequency of false pulse detection generated in total darkness, due to thermal noise or tunneling and not photons. It is measured in counts per second (cps, or Hz). Room

temperature Si-SPADs show 10–100 kcps, while cooled, optimized detectors often achieve much lower values (e.g., <1 cps). It is crucial for determining the signal-to-noise ratio in low-light applications.

c) Timing jitter : Timing jitter in a single-photon detector (SPD) is the difference in time between the actual arrival of a photon and the generation of the signal, typically measured in picoseconds (ps), with SNSPDs having 3 ps - 50 ps and SPADs from 10-100 ps.

d) Recovery time (Dead time): It is the interval required for the device to reset and regain sensitivity after detecting a photon. It typically ranges from tens of nanoseconds for Superconducting Nanowire Single-Photon Detectors (SNSPDs) to microseconds for InGaAs Avalanche Photodiodes. It sets the limit of bit rate in applications like quantum key distribution.

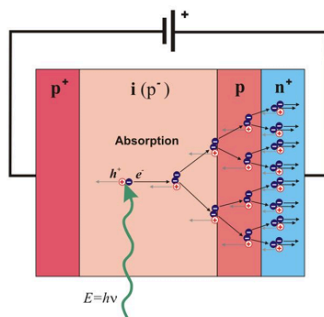
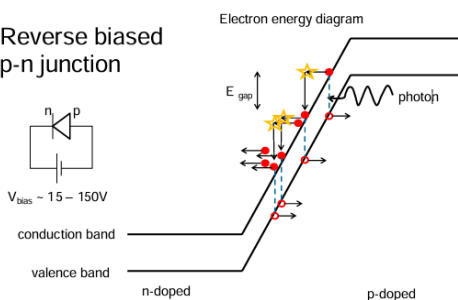
(a) Single Photon Avalanche Diode (SPAD)

Working Principle : When an APD operates in **Geiger mode** (reverse biased above breakdown voltage), the generated electron-hole pair triggers an avalanche reverse current. The spike in the reverse current signals the detection of a single photon. The spiked reverse current needs to be quenched to prepare the SPAD for the next photon. This is done in two ways: a) By using a current limiting resistance (Passive quenching) or b) by lowering the reverse bias voltage to below the breakdown voltage and restoring it to Geiger mode after the dead time (active quenching). The jitter time varies from 40 ps -200 ps.

Features : High sensitivity, Room temperature operation, needs active quenching. It has applications in Quantum key distribution, LIDAR,

Fluorescence imaging.

- Reverse biased p-n junction



Electron band energy diagram in a reverse biased p-n junction. An absorbed photon creates an electron (solid circle) and a hole (empty circle). The large reverse bias voltage produces an electric field in the junction that accelerates the electron towards the n doped side. When an electron collides with an atom, its kinetic energy can be used to promote an extra electron into the conduction band, thus triggering the avalanche.

Errors Associated with Avalanche Photodetectors and SPADs:

Afterpulsing: After a photon initiates an avalanche, it is possible for an electron in an excited state to gain enough thermal energy to reach the conduction band and trigger a second avalanche at a very short time interval. This phenomenon is called afterpulsing.

Dead time and pulse widths By varying the value of the resistor in series with the SPAD, the RC time constant of the circuit can be controlled ($\tau = RC$). This controls the dead time of the detector (time when the detector cannot register the arrival of the next photon).

Superconducting Nanowire Single Photon Detector (SNSPD)

Construction : It is made of a thin and narrow superconducting wire arranged in a Meander Geometry with very small spacing between the meandering wires. The wire is kept at a temperature below its superconducting transition temperature making the entire wire superconducting and is connected to a DC source. A small resistance is kept in series which regulates the biasing current passing through the SNSPD. The biasing current is maintained just below the critical current of the wire. The entire arrangement is connected to a special type of electronic circuit called the **“Readout”**. When a photon of energy higher than the superconducting gap hits the meander, locally the Cooper pairs are broken and the local critical current decreases. If the critical current decreases below the bias current then the superconductivity is destroyed and it turns into a non-superconducting hot spot (normal metal) locally. The resistance of the hotspot becomes higher than the Readout resistance (about 1 k Ω), which is in parallel to the hot spot. Hence the bias current now flows through the Readout. The Readout is effectively a shunt resistance for the hotspot region. The Readout produces a measurable voltage pulse for every single photon that strikes the SNSPD. By measuring the voltage we can measure the wavelength of the incident photon. When the current passes through the shunt resistance, the hot-spot region cools rapidly and returns to the superconducting state, thus making it ready to detect a second photon.

The readout system is critical for converting the tiny, high-speed resistance change caused by a single photon absorption into a recordable electrical or optical pulse. The readout circuit is responsible for biasing the detector, quenching the hotspot, restoring the superconducting state, and amplifying the signal for external recording while minimizing noise and heat load.

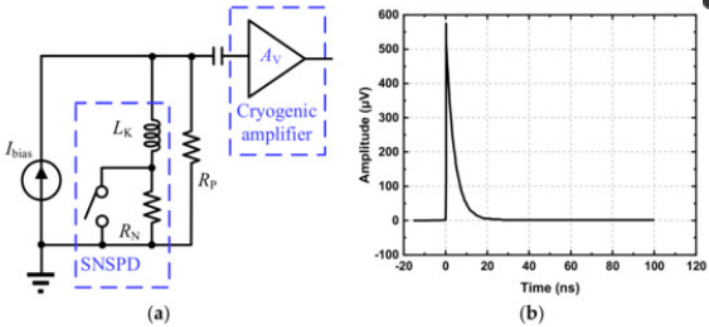


Figure: (a) SNSPD model and its connection with the cryogenic amplifier; (b) Typical SNSPD output pulse signal.

With a current bias, the working behavior of the SNSPD can be described with an equivalent electrical lumped model. [An electrical lumped model simplifies a complex, physically distributed electrical circuit into a network of finite idealized components—resistors, capacitors, and inductors]. The SNSPD consists of a kinetic inductor L_K which is functionally and physically the same as the inductive term derived from the superfluid component of the two-fluid model of superconductivity, a resistor R_N (representing the time-dependent, non-linear resistance of the hot spot created when a photon is absorbed), and a switch. To establish a proper current loop and to quench the device, a parallel resistor R_P of 50Ω is added. L_K is much larger than the Faraday inductance which is ignored along with the parasitic capacitance of SNSPD. When the SNSPD is in the superconducting state, the switch is closed and R_N is zero (shorted by switch). When an incident photon is absorbed by the SNSPD, the switch is opened, the current flows through R_N which being very high (order of Mega ohms) pushes the current into the parallel resistor R_P . Accordingly, a pulse voltage signal, typically with several hundred microvolts amplitude, is generated, as shown in **Figure b**. Normally, its rising time is around 200 ps, while its falling time ranges from 10 to 60 ns. The superconducting nanowire is biased close to its critical

current. Photon absorption creates a **localized resistive hotspot**, producing a detectable voltage pulse. It has a very high detection efficiency ($>90\%$), ultra-low dark counts (1 cps), Excellent timing resolution (~ 10 ps)

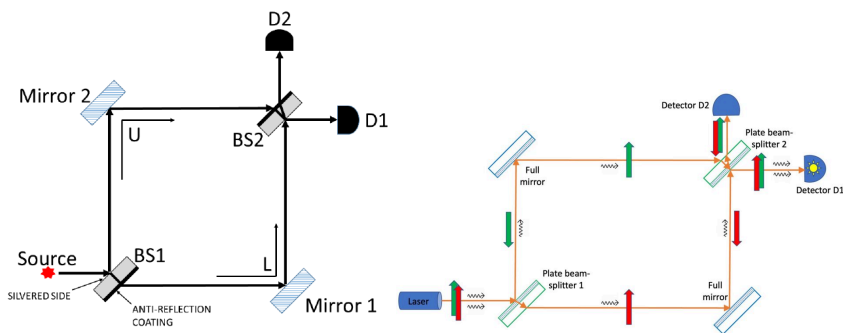
Limitations : Requires **cryogenic temperatures (2–4 K)**

Applications : Quantum optics, Astronomical photon detection, Quantum computing readout

Mach-Zehnder Interferometer (MZI)

The **Mach-Zehnder Interferometer (MZI)** is a beam division interferometer used to measure **phase differences** between two coherent light beams. It is extremely configurable compared to other interferometers. It is an optical device used to split light into two paths, then recombine them to measure phase shifts caused by variations in refractive index, concentration, or temperature. It acts as a switch, modulator, or sensor in integrated optics, utilizing interference to control output. It finds application in quantum sensing, cryptography, computing and entanglement.

Construction and Working: It has a monochromatic source S (usually a Laser source), two beam splitters BS-1 and BS-2, two plane mirrors M1 and M2 and two detectors D1 and D2 arranged as shown in the figure below.



Structure: Two beam splitters (BS1 and BS2), two mirrors (M1 and M2), two optical paths (L and U) and two output ports (D1 and D2)

Working Principle

1. Incident light is split into two paths and the beams are recombined at the second beam splitter.
2. Outcome at BS2 depends on relative phase difference between the two beams. Accordingly detection happens in any one of the two detectors D1 or D2 wherever constructive interference occurs.

Key Features

1. Extremely sensitive to phase changes
2. Basis for optical modulators and sensors

Applications

3. Electro-optic modulation
4. Refractive index sensing
5. Optical coherence tomography
6. Quantum interference experiments

The second figure above and the table below gives a list of how different processes alter the phase of the light beam.

	Initially in medium with lower n	Initially in medium with higher n
Reflection at interface	Phase shift of π	No phase shift
Transmission at interface	No phase shift	No phase shift
Propagation through a medium	Phase shift ϕ depends on thickness and refractive index n of the medium	

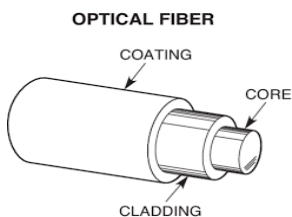
It can be seen that the upper beam U does not reflect from the front face of the second beam splitter BS2 due to the presence of anti reflection coating on the front face. Hence it goes inside the beam splitter and reflects from the back side at the glass-air interface. Due to Fresnel's law, this accrues no phase change.

The net result is that after recombination and interference at the

second beam splitter, the entire laser light is detected at D0 and none at D1, due to constructive interference in the former and destructive interference at the latter detector.

OPTICAL FIBERS

Optical fibres are wave guides used in optical communications. They are thin, cylindrical, transparent flexible dielectric fibres. The working structure of optical fibre consists of three layers. The inner cylinder which is made of glass or plastic is called the Core, while the surrounding layer which envelops the inner core is called Cladding. It has a slightly lower refractive index than core. The core and the cladding layers are enclosed in a polyurethane jacket called sheath which safeguards the working structure of fibre against chemical reactions, mechanical abrasion and crushing etc.



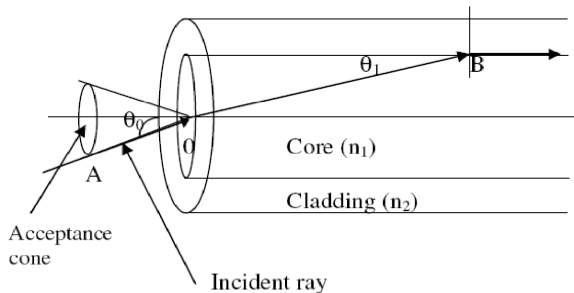
Propagation mechanism in Optical fibre:

If the light ray which enters into the core strikes the core and cladding interface at angles greater than critical angle for the two materials, the light ray undergoes multiple total internal reflections within the fibre core, thereby preserving the signal strength.

Numerical Aperture and Ray Propagation in the Fibre:

Consider an optical fibre consists of core and cladding material placed in air medium. Let n_0 , n_1 and n_2 be the refractive indices of surrounding air medium, core and cladding material respectively. The RI of cladding is always lesser than that of core material ($n_2 < n_1$) so that the light rays propagate through the fibre. Let us consider the

special case of ray which suffers critical incident at the core cladding interface. The ray travels along AO entering into the core at an angle of θ_0 with respect to the fibre axis. Let it be refracted along OB at an angle θ_1 in the core and further proceed to fall at critical angle of incidence = $90 - \theta_1$ at B on the interface of core and cladding. For critical angle of incidence, the refracted ray grazes along the interface.



It is clear from the figure that a ray that enters at an angle of incidence less than θ_0 at O, will have to be incident at an angle greater than the critical angle at the core-cladding interface, and gets total internal reflection in the core material. When OA is rotated around the optical axis, it describes an acceptance cone. Rays within this cone will only be totally internally reflected and propagate through the fibre. The angle θ_0 is called the wave guide acceptance angle or the acceptance cone half-angle and **$\sin \theta_0$ is called the numerical aperture (N.A.)** of the fibre. It determines the light gathering ability of the fibre.

Let n_0 , n_1 and n_2 be the refractive indices of surrounding medium, core and cladding respectively for the given optical fibre. By applying the Snell's law at O,

$$n_0 \sin \theta_0 = n_1 \sin \theta_1$$

$$\sin \theta_0 = \frac{n_1}{n_0} \sin \theta_1 \text{-----(1)}$$

At the point B on the core and cladding interface, the angle of incidence = $90 - \theta_1$

Applying Snell's law at B

$$n_1 \sin(90 - \theta_1) = n_2 \sin 90$$

or $n_1 \cos \theta_1 = n_2$

$$\cos \theta_1 = \frac{n_2}{n_1} \text{-----(2)}$$

From equation (1)

$$\sin \theta_0 = \frac{n_1}{n_0} \sin \theta_1 = \frac{n_1}{n_0} (\sqrt{1 - \cos^2 \theta_1})$$

$$\sin \theta_0 = \frac{n_1}{n_0} \sqrt{1 - \frac{n_2^2}{n_1^2}} = \frac{n_1}{n_0} \sqrt{\frac{n_1^2 - n_2^2}{n_1^2}} = \frac{\sqrt{n_1^2 - n_2^2}}{n_0}$$

If the medium surrounding the fibre is air, then $n_0 = 1$,

Therefore, $\sin \theta_0 = \sqrt{n_1^2 - n_2^2}$
 $\sin \theta_0 = \text{Numerical aperture: NA}$

Therefore, $NA = \sin \theta_0 = \sqrt{n_1^2 - n_2^2}$

If θ_i is the angle of incidence of an incident ray, then the ray will be able to propagate,

If $\theta_i < \theta_0$

or $\sin \theta_i < \sin \theta_0$

or $\sin \theta_i < NA$

Fraction Index Change (Δ): It is the ratio of the difference in the refractive indices between the core and the cladding to the refractive

index of core of an optical fibre. It is also known as relative core clad index difference.

$$\Delta = \frac{n_1 - n_2}{n_1}$$

Relation between NA and Δ

We have RI change $\Delta = \frac{n_1 - n_2}{n_1}$

Or $\Delta n_1 = (n_1 - n_2) \text{-----(1)}$

We know that,

$$\begin{aligned} N.A. &= \sqrt{n_1^2 - n_2^2} \\ &= \sqrt{(n_1 + n_2)(n_1 - n_2)} \\ &= \sqrt{(n_1 + n_2)n_1\Delta} \end{aligned}$$

Since $n_1 \sim n_2$, $(n_1 + n_2) = 2n_1$

$$N.A. = \sqrt{2n_1^2\Delta}$$

Therefore, $N.A. = n_1\sqrt{2\Delta}$

NA can be increased by increasing Δ , so as to receive maximum light into the fibre. However, fibres with large Δ will not be useful for optical communication due to the occurrence of a phenomenon inside the fibre called multipath or intermodal dispersion. It introduces a time delay factor causing distortion of the transmitted optical signal and pulse broadening, limiting the communication distance.

Modes of Propagation:

The possible number of paths of light in an optical fibre determines the number of modes available in it. It also determines the number of independent paths for light that a fibre can support for its propagation without interference and mixing.

We may have a single mode fibre supporting only one signal at a time or multimode fibre supporting many rays at a time.

Such number of modes supported for propagation in the fibre is determined by a parameter called V-number. If the surrounding medium is air then the V- number is given by

$$V = \frac{\pi d}{\lambda} \sqrt{(n_1^2 - n_2^2)}$$

$$V = \frac{\pi d}{\lambda} (NA)$$

or

where d is the core diameter, n_1 is the refractive index of core, n_2 is the refractive index of the cladding and λ is the wavelength of the light propagating through the fibre.

$$V = \frac{\pi 2r}{\lambda} \sqrt{n_1^2 - n_2^2} = \frac{(r)2\pi}{\lambda} \sqrt{n_1^2 - n_2^2} = r \cdot k \sqrt{n_1^2 - n_2^2}.$$

Where k is the propagation constant ($k = 2\pi/\lambda$)

If fibre is surrounded by a medium of refractive index n_0 , then

$$V = \frac{\pi d}{\lambda} \frac{\sqrt{n_1^2 - n_2^2}}{n_0}$$

For step index fibre, the number of modes $\approx \sqrt{V^2}/2$

For graded index fibre, the number of modes $\approx V^2/4$

Types of Optical fibres:

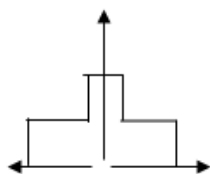
The optical fibres are classified under 3 categories. They are

- a) Step index Single mode fibre (SMF)
- b) Step index multi mode fibre (MMF)
- c) Graded index Multi Mode Fibre (GRIN)

This classification is done depending on the refractive index profile and the number of modes that the fibre can guide.

Refractive Index Profile (RI):

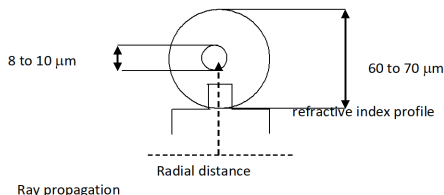
Generally, the refractive index of cladding material is always constant throughout the fibre. But the core material refractive index may either remain constant or subjected to variation in a particular way. This variation of RI of core and cladding materials with respect to the radial distance from the axis of the fibre is called refractive index profile. This can be represented as follows,



RI profile of Step index fibre

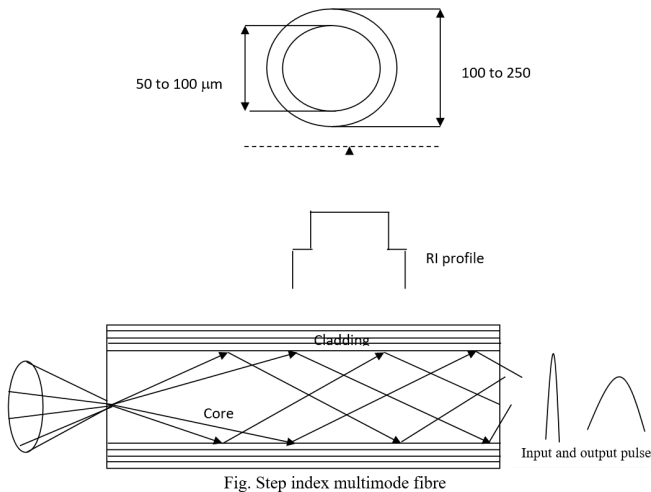
X axis: Radial distance from the centre of the fibre

Step index Single mode fibre (SMF): It has uniform refractive index for both core and cladding. The core-cladding interface discontinuity causes the RI profile to take the shape of a step. The core diameter is about 8 to 10 μm and the cladding diameter is 60 to 70 μm . Because of its narrow core, it can guide just a single mode as shown in Figure. It is called single mode fibre and has zero Intermodal dispersion. They constitute 80% of all the fibres that are manufactured in the world today. They need lasers as the source of light. Though less expensive, it is very difficult to join the optical fibres. They find particular applications in submarine cable system.



Step index multimode fibre (MMF):

The construction and refractive index profile of a MMF is similar to that of a SMF but for the difference that **its core has a much larger diameter**. Hence MMF supports propagation of a large number of modes as shown in the figure. The step-index multimode fibre can accept either diode laser or LED (light emitting diode) as source of light. It is the least expensive of all. Since multi modes are propagating through this fibre with different paths, intermodal dispersion is maximum in this fibre. Its typical application is in data links which has lower bandwidth requirements.



Graded index multimode fibre (GRIN)

The construction of Graded index fibre is similar to that of multimode step-index fibre, except for the refractive index of the core. The refractive index of the core varies across the core diameter (radially graded) as shown in figure, while the refractive index of the cladding is fixed. In this fiber, a number of modes can be transmitted. The rays move in a sinusoidal path through the core. Light travels at lower speed in the high-index region. Since the fastest components of the ray take the longer path and the slower components take the shorter path in the core, the travel time of the different modes will be almost the same. This reduces the effect of intermodal dispersion and hence losses are minimal with little pulse broadening. These fibres are most suitable for

large bandwidth, medium distance and medium bit rate communication systems. For such cables, either a laser or LED source can be used to couple the signal into the core.

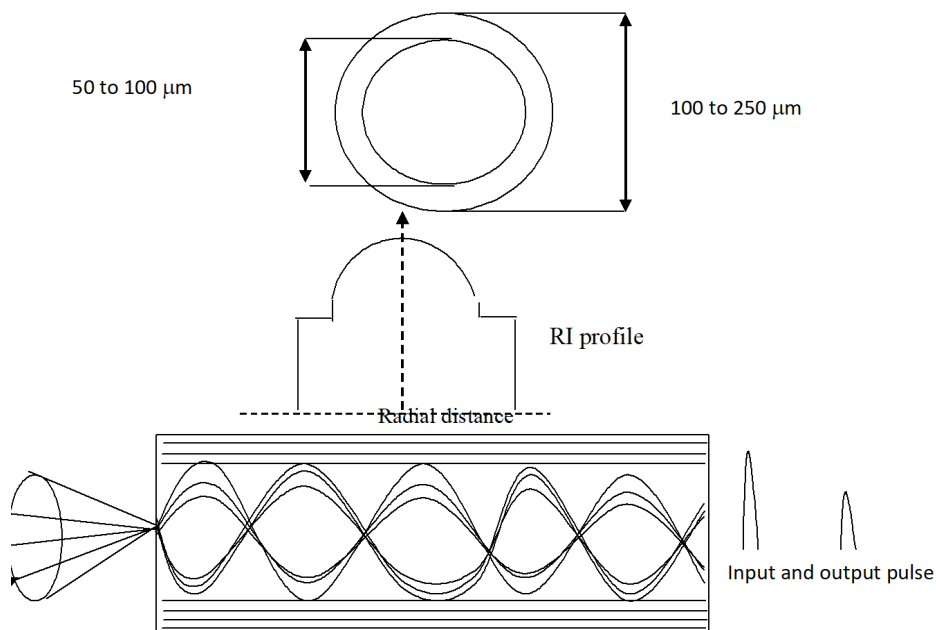


Fig. Graded index multimode fibre

Differences between single and multimode fibres:

Single mode fibre	Multi modefibre
● Only one mode propagates ● Smaller core diameter ● Low dispersion of signal ● works for longer distances ● Launching of light and connecting two fibres are difficult	● multiple modes can propagate ● Larger core diameter ● More dispersion of signal ● works for shorter distances only ● Launching of light and connecting of fibres is easy

Differences between step and graded index fibres:

Step index fibre	Graded index fibre
- Refractive index of core is uniform - Propagation of light is in the form of meridional rays - Step index fibres has lower bandwidth - Distortion is more (in multimode) - No. of modes for propagation $N_{\text{step}} = V^2/2$	- Refractive index of core is not uniform - Propagation of light is in the form of skew rays - Graded index fibres has higher bandwidth - Distortion is less - No. of modes for propagation $N_{\text{grad}} = V^2/4$

Attenuation in optical fibres:

The total energy loss suffered by the signal due to the transmission of light in the fibre is called **attenuation**.

The important factors contributing to the attenuation in optical fibre are
i) Absorption loss ii) Scattering loss iii) Bending loss
iv) Intermodal dispersion loss and v) coupling loss.

Attenuation is measured in terms of attenuation co-efficient and it is the loss per unit length. It is denoted by the symbol α . Mathematically attenuation of the fibre is given by,

$$\alpha = -\frac{10 \times \log_{10} \left(\frac{P_{\text{out}}}{P_{\text{In}}} \right)}{L} \text{ dB/km}$$

Where P_{out} and P_{in} are the power output and power input respectively, and L is the length of the fibre in km.

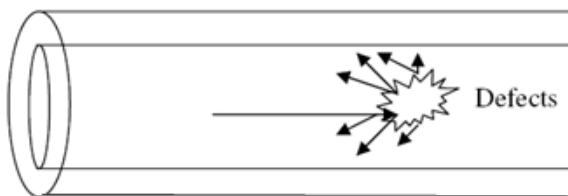
Therefore, Loss in the optical fibre = $\alpha \times L$

1. Absorption loss:

There are two types of absorption: (a) extrinsic impurities, (b) Intrinsic absorption. Extrinsic impurities are generally transition metal ions such as iron, chromium, cobalt, copper. These impurities electronically absorb the photons and get excited to higher energy levels. Later, the absorbed energy is released as either heat energy or light which is in a different wavelength and out of phase with respect to the signal. The other impurity which would cause significant absorption loss is the OH^- (Hydroxyl) ion, which enters into the fibre constitution at the time of fibre fabrication. **Intrinsic absorption** is absorption by the material constituting the fibre itself. Assuming that there are no impurities and the material is free of all inhomogeneities, this sets the lowest limit on absorption for a given material.

2. Scattering loss:

The signal power loss occurs due to the scattering of light energy from imperfections and defects, which are of molecular size, present in the fibre itself. This scattering is inversely proportional to the fourth power of the wavelength of the light transmitted through the fibre and is called Rayleigh scattering. The loss can be minimized by transmitting light of large wavelengths.



3. Bending losses (radiation losses):

There are two types of bending losses in optical fibre a) macroscopic and b) microscopic bending loss.

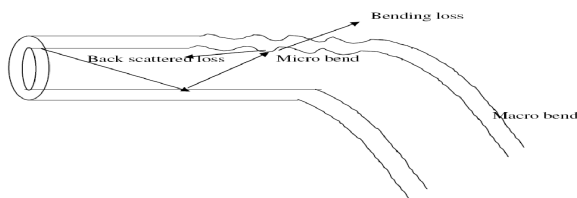
a) Macroscopic bends:

Macroscopic bends occur due to the wrapping of fibre on a spool or turning it around a corner. The loss will be negligible for small bends but increases rapidly until the bending reaches a certain critical radius of curvature. If the fibre is too bent, then there is a possibility of the

light ray escaping through the cladding material without undergoing any total internal reflection at the core-cladding interface.

b) Microscopic bends:

This refers to repetitive small-scale variations from linearity of the fibre axis, due to non-uniformities in the manufacturing and lateral pressures created during the cabling of the fibre. These microscopic bends cause irregular reflections at the core-cladding interface and some of them reflect back or leak through the fibre. This loss could be minimized by extruding a compressible sheath over the fibre which can withstand the stresses while keeping the fibre relatively straight.



4. Coupling losses:

Coupling losses occur when the ends of the fibres are connected. At the junction of coupling, air film may exist or joint may be inclined or may be mismatched and they can be minimized by following the technique called splicing.